Mean-Pooled Cosine Similarity is Not Length-Invariant: Theory and Cross-Domain Evidence for a Length-Invariant Alternative

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Abstract

Mean-pooled cosine similarity is the default metric for comparing neural representations across languages, modalities, and tasks. We establish that this metric is *not* length-invariant: under the anisotropy that characterises modern transformer representations, mean-pooled cosine grows monotonically in sequence length, independent of representational content. We prove this from first principles, validate it on random vectors with no model involvement, and demonstrate it empirically across code, natural-language, and vision domains. On HumanEvalPack across four code LLMs, length ratio alone explains $R^2 = 0.52$–0.75 of cross-language "Python proximity," while AST depth and shared-token fraction add less than 3%. Substituting linear CKA reduces explained variance by 83% and *reverses* the sign of the length coefficient ($\beta_{\text{len}} : +0.86 \rightarrow -0.37$). The same pattern holds in Mistral-7B on WMT parallel pairs (R^2=0.23–0.33 for cosine; R^2<0.01 for CKA). We argue that length-invariant metrics such as CKA should be the default for cross-representation comparisons, and that recent claims of cross-lingual representational convergence built on mean-pooled cosine warrant re-examination.

1 Introduction

Comparing how a neural network represents different inputs is foundational to mechanistic interpretability. The standard procedure averages token-level hidden states into a single vector and reports cosine similarity. This mean-pooled cosine has become the de facto metric underpinning claims that multilingual LLMs "think in English" (Schut et al., 2025; Wendler et al., 2024)—a pivot-language pattern recently extended to code models, whose concept space aligns most closely with English keywords (Kargaran et al., 2025a). Python, the most token-compact and training-dominant programming language, invites an analogous within-code question that we examine here.

We show that the metric is not length-invariant. Under anisotropic representations (Ethayarajh, 2019; Mu et al., 2018; Gao et al., 2019)—the regime all modern transformers operate in—mean-pooled cosine grows monotonically with sequence length regardless of content. The mechanism is a $1/\sqrt{n}$ concentration of pooled vectors toward the shared anisotropy direction. This is large enough to dominate published cross-language similarity analyses.

Key findings.

F1. Mean-pooled cosine is monotonically increasing in sequence length under anisotropy (Prop. 1), validated by a synthetic experiment on random vectors with no model involvement.

F2. Across four code LLMs and 164 problems, length ratio alone explains 52–75% of Python-proximity variance. AST depth and shared tokens add less than 3%.

F3. Substituting CKA on the same data reduces explained variance by 83% and flips the sign of the length coefficient. The metric-level conclusion reverses.

F4. The artifact is not code-specific: Mistral-7B on WMT EN–FR/DE shows R^2=0.23–0.33; CKA on the same data has R^2<0.01.

F5. In CLIP ViT-B/32, mean-pooling *reduces* the length effect vs. EOS-pooling (R^2<0.01 vs. 0.21), as predicted: CLIP's contrastive head suppresses anisotropy.

What we are and are not claiming. We are *not* claiming representational convergence is illusory. Our positive controls show genuine convergence is large and detectable with CKA. We are claiming that mean-pooled cosine cannot distinguish genuine convergence from a tokenizer length differential, and that the reference languages—English in prior cross-lingual work, Python in the within-code analysis we examine—are precisely those with systematically shorter tokenisations.

2 Related Work

Wendler et al. (2024) report Llama-2 representations cluster around an English pivot using mean-pooled cosine. Schut et al. (2025) extend this with logit-lens but treat cosine as the primary metric. Kargaran et al. (2025a) apply logit-lens and position-weighted cosine to code LLMs, finding the concept space aligns with English keywords; Kargaran et al. (2025b) do the same for low-resource languages. None control for sequence length. Petrov et al. (2023) document systematic tokenisation count differences across languages—the direct input to our artifact. Ethayarajh (2019) demonstrated transformer anisotropy; Gao et al. (2019) and Mu et al. (2018) proposed mitigations. Kornblith et al. (2019) introduced linear CKA as a length- and rotation-invariant similarity measure; Robert and Escoufier (1976) is its statistical antecedent; Davari et al. (2023) discuss adversarial failure modes. Conneau et al. (2020) established the multilingual pivot hypothesis; our results imply recent metric-based evidence for it may be confounded.

3 Mechanism

3.1 Anisotropy and mean pooling

Modern transformer hidden states are anisotropic: two random states from the same model often have cosine similarity > 0.5 (Ethayarajh, 2019). Writing each token state as

$$h_i = \mu + \sigma\epsilon_i, \qquad \|\mu\| \gg \sigma\sqrt{d}, \tag{1}$$

where μ is a layer-specific shared direction and ϵ_i is zero-mean noise, mean pooling over n tokens gives

$$\bar{h} = \mu + \frac{\sigma}{\sqrt{n}}\,\bar{\epsilon}. \tag{2}$$

The shared direction is preserved exactly; the noise shrinks as $1/\sqrt{n}$. Longer sequences therefore lie closer to μ, and consequently closer (in cosine) to any pooled vector from the same distribution.

3.2 Formal result

Proposition 1 (Length dependence of mean-pooled cosine)**.** *Let $\{x_1, \ldots, x_m\}$ and $\{y_1, \ldots, y_n\}$ be independent i.i.d. sequences in $\mathbb{R}^d$ with mean $\mu \neq 0$ and covariance $\sigma^2 I_d$. For $d \gg 1$ and $\|\mu\|^2 = \Theta(d\mu_{\text{comp}}^2)$,*

$$\mathbb{E}[\cos(\bar{x}, \bar{y})] \approx \frac{1}{\sqrt{1 + \frac{\sigma^2 d}{m\|\mu\|^2}}\sqrt{1 + \frac{\sigma^2 d}{n\|\mu\|^2}}}, \tag{3}$$

which is strictly increasing in both m and n.

Proof sketch. By the CLT, $\bar{x} = \mu + (\sigma/\sqrt{m})\epsilon_x$ with $\epsilon_x \sim \mathcal{N}(0, I_d)$.

Numerator. $\langle \bar{x}, \bar{y} \rangle = \|\mu\|^2 + (\sigma/\sqrt{m})\mu^\top \epsilon_x + (\sigma/\sqrt{n})\mu^\top \epsilon_y + (\sigma^2/\sqrt{mn})\epsilon_x^\top \epsilon_y$. The first two cross-terms have mean zero. In high dimensions $\epsilon_x^\top \epsilon_y = \mathcal{O}(\sqrt{d})$ while $\|\mu\|^2 = \Theta(d\mu_{\text{comp}}^2)$, so the cross-noise term is dominated and $\mathbb{E}\langle \bar{x}, \bar{y} \rangle = \|\mu\|^2$.

Denominator. $\mathbb{E}\|\bar{x}\|^2 = \|\mu\|^2 + \sigma^2 d/m$ (since $\|\epsilon_x\|^2$ concentrates around d); analogously for $\bar{y}$.

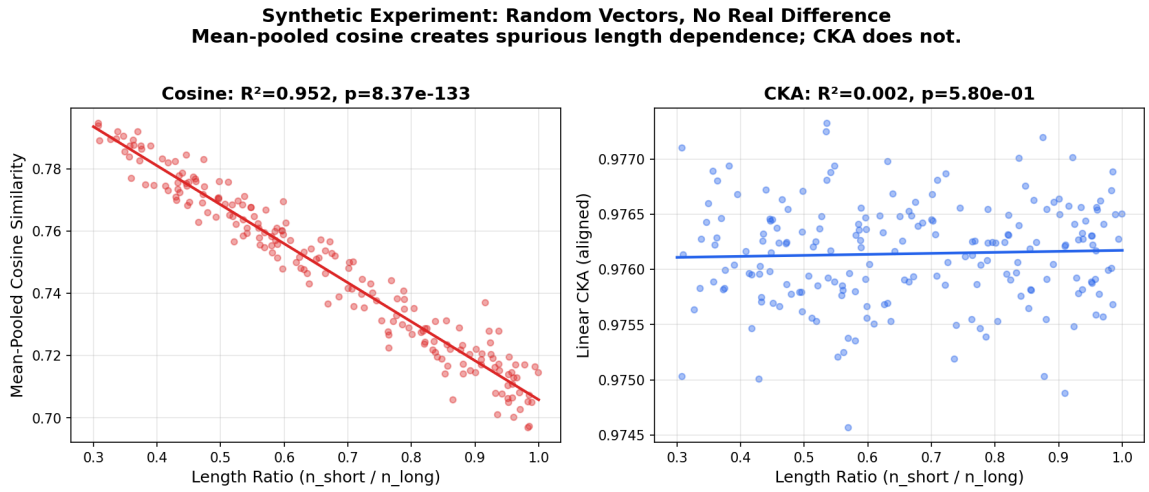


Figure 1: **Synthetic validation.** 200 random-vector pairs in $\mathbb{R}^{4096}$. Mean-pooled cosine (left) tracks length ratio as Eq. (3) predicts. CKA on aligned subsets (right) does not. No model or semantics involved.

Assembly. $\mathbb{E}\cos(\bar{x}, \bar{y}) \approx \|\mu\|^2 / \sqrt{(\|\mu\|^2 + \sigma^2 d/m)(\|\mu\|^2 + \sigma^2 d/n)}$, which simplifies to Eq. (3). Since $k \mapsto \sqrt{1 + c/k}$ is decreasing in k for $c > 0$, the denominator decreases in both m and n, so the cosine is increasing in both. □

The size of the effect is governed by $\rho = \sigma^2 d / \|\mu\|^2$. Taylor expanding, $\mathbb{E}\cos \approx 1 - \frac{1}{2}\rho(1/m + 1/n)$: the artifact is largest when lengths are short and asymmetric, and scales linearly with anisotropy.

3.3 Why CKA is immune

Linear CKA (Kornblith et al., 2019) compares full token-level matrices $X \in \mathbb{R}^{n_X \times d}$, $Y \in \mathbb{R}^{n_Y \times d}$ via

$$\mathrm{CKA}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \cdot \|Y^\top Y\|_F}. \tag{4}$$

There is no pooling step, so the $1/\sqrt{n}$ concentration does not arise. CKA is also invariant under invertible linear transformations of the column space—exactly what shifting along μ amounts to.

3.4 Synthetic validation

To verify the mechanism without any model, we generate 200 pairs of random vectors in $\mathbb{R}^{4096}$ from $h_i = \mu + \epsilon_i$ with $\|\mu\| = 10$, $\sigma = 1$, at varying length ratios $r \sim \mathrm{Uniform}(0.3, 1.0)$.

Figure 1 confirms that mean-pooled cosine tracks the length ratio exactly as predicted, while CKA is flat. This rules out any explanation involving model behaviour, language structure, or content semantics.

4 Experimental Setup

Models. *Code:* CodeLlama-7B, CodeLlama-7B-Python, CodeLlama-13B (Rozière et al., 2023), Qwen2.5-Coder-7B (Hui et al., 2024). *NLP:* Mistral-7B-v0.1. *Vision:* CLIP ViT-B/32 (Radford et al., 2021).

Datasets. *Code:* HumanEvalPack (Muennighoff et al., 2024), 164 problems with parallel Python/Java/JavaScript/Go solutions. *NLP:* WMT14 EN–FR (n=442) and WMT16 EN–DE (n=428), filtered for ≥ 3 shared surface-form tokens for position-aligned CKA. *Vision:* 400 synthetic captions of varying length paired with a fixed image through CLIP.

Metrics. (i) *Mean-pooled cosine*: $\cos(\bar{h}_A, \bar{h}_B)$, averaging over middle layers ($n/4$ to $3n/4$). (ii) *Linear CKA* (Kornblith et al., 2019) on aligned token positions. (iii) *RV coefficient* (Robert and Escoufier, 1976), the matrix generalisation of squared correlation.

Confounds. We regress on: *length ratio* ($\min / \max$ token counts), *AST depth range*, and *shared-token fraction*. All variables standardised; β are standardised weights.

5 Results

5.1 Length explains nearly all of Python proximity

Table 1 shows that in every model, length ratio dominates with β=0.74–0.88 and $p<10^{-27}$, explaining R^2=0.52–0.75. AST depth contributes at most $R^2 \leq 0.09$; its univariate negative correlation reverses to near-zero in the multivariate model (Simpson's paradox: depth covaries with length). Adding all confounds beyond length adds less than 1% of explained variance in three of four models.

Table 1: Multiple regression of Python proximity (mean-pooled cosine) on three confounds. R^2_ℓ: length-only R^2 with 95% bootstrap CI (B=5000); n=164 per model.

Model	R^2_ℓ	95% CI	β_{len}	β_{depth}
CL-7B	0.72	[0.62, 0.80]	0.86	0.03
CL-7B-Py	0.75	[0.66, 0.81]	0.88	0.05
CL-13B	0.67	[0.54, 0.76]	0.84	0.07
Qwen-7B	0.52	[0.36, 0.67]	0.74	0.05

5.2 CKA reverses the sign of the conclusion

Table 2 compares all three metrics on CodeLlama-7B. CKA reduces explained variance by 83% and reverses the sign of β_{len}. The substantive conclusion—Python is a hub vs. Python is an outlier—depends entirely on which metric is used.

Table 2: Metric comparison on CodeLlama-7B (n=164). The sign of β_{len} flips between cosine and CKA.

Metric	R^2_{full}	β_{len}	r_{univ}
Cosine	0.72	+0.86	+0.85
RV coefficient	0.30	+0.51	+0.53
Linear CKA	0.13	−0.37	−0.35

5.3 Generalisation to natural language

Figure 2 shows the artifact in Mistral-7B on English–French. The German result is larger (R^2=0.33), consistent with German's longer relative tokenisation; CKA on the same data has R^2<0.006. See Appendix for the EN–DE plot.

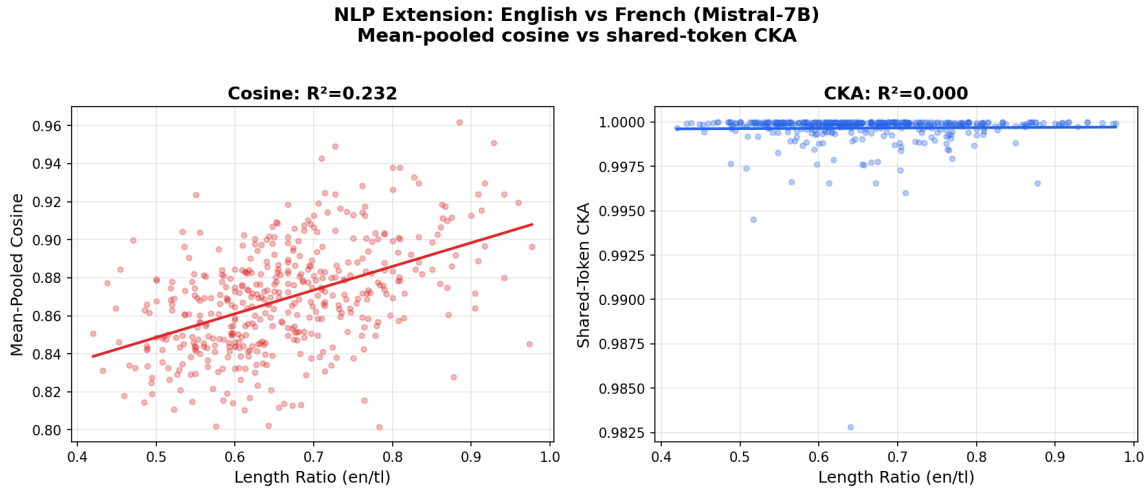


Figure 2: **NLP: English–French (Mistral-7B, n=442).** Cosine correlates with length ratio at R^2=0.23 ($p<10^{-26}$); CKA shows no dependence ($R^2<0.001$, p=0.69).

5.4 Vision: architecture-dependent suppression in CLIP

In CLIP ViT-B/32, standard EOS-pooled cosine shows R^2=0.21; the non-standard mean-pooled variant shows $R^2<0.01$. Mean-pooling *reduces* the artifact because CLIP's contrastive head produces lower-anisotropy embeddings, removing the substrate the mechanism requires. This cross-domain pattern—large in code LLMs, intermediate in Mistral, suppressed in CLIP—tracks anisotropy ρ exactly as the theory predicts. Full CLIP and EN–DE figures are in the Appendix.

5.5 What CKA shows about genuine convergence

Raw mean CKA at matched positions is large: 0.9997 for Mistral-7B on EN–FR/DE; 0.997 for CodeLlama-13B and 0.913 for Qwen2.5-Coder-7B; 0.65 for CodeLlama-7B on Python vs. {Java, JS, Go}. Genuine cross-language convergence is real and in NLP essentially saturating. What does not survive metric correction is the asymmetric privileged-pivot framing: under CKA, β_{len} flips sign (Tab. 2) and the residual structure is symmetric.

6 Discussion and Conclusion

Implications. Wendler et al. (2024), Schut et al. (2025), and Kargaran et al. (2025a) use mean-pooled cosine as primary evidence for pivot-language claims. In each case the reference language has systematically shorter tokenisations. On the strength of the metric alone, the data are equally consistent with genuine convergence and a pure tokeniser-length differential. Convergence is real; only the privileged-pivot framing was metric-induced.

Recommendations. (i) Use a length-invariant metric (linear CKA, RV) whenever input lengths can vary. (ii) Report length statistics alongside any similarity result. (iii) Validate findings with multiple metrics: a result that holds under cosine but vanishes under CKA should be presumed a metric artifact. (iv) If anisotropy mitigation (Mu et al., 2018) is acceptable, it directly attenuates the artifact by reducing ρ.

Limitations. Davari et al. (2023) show CKA can be misled under adversarial column-scaling, and our shared-surface-form alignment introduces a selection effect. We test only linear CKA; kernel variants may

differ. Eq. (3) assumes isotropic noise (approximate); the qualitative monotonic length-dependence is what we test and observe.

Conclusion. Mean-pooled cosine is not length-invariant under transformer anisotropy. We prove this from first principles, validate it on pure random vectors, and demonstrate it across code, NLP, and vision domains. Substituting linear CKA on identical data cuts length-explained variance by 83% and flips the sign of β_{len} in code, while confirming that genuine cross-language convergence is large. The convergence was real; only the asymmetric pivot framing was metric-induced. Length-invariant metrics should be the default for cross-representation comparisons.

References

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020. URL https://arxiv.org/abs/1911.02116.

MohammadReza Davari, Stefan Horoi, Amine Natik, Guillaume Lajoie, Guy Wolf, and Eugene Belilovsky. Reliability of CKA as a similarity measure in deep learning. In *International Conference on Learning Representations (ICLR)*, 2023. URL https://arxiv.org/abs/2210.16156.

Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 55–65, 2019. URL https://arxiv.org/abs/1909.00512.

Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-Yan Liu. Representation degeneration problem in training natural language generation models. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://arxiv.org/abs/1907.12009.

Binyuan Hui, Jian Yang, Zeyu Cui, et al. Qwen2.5-coder technical report. 2024. URL https://arxiv.org/abs/2409.12186.

Amir Hossein Kargaran, Yihong Liu, François Yvon, and Hinrich Schütze. How programming concepts and neurons are shared in code language models. In *Findings of the Association for Computational Linguistics (ACL)*, 2025a. URL https://arxiv.org/abs/2506.01074.

Amir Hossein Kargaran, Ali Modarressi, Nafiseh Nikeghbal, Jana Diesner, François Yvon, and Hinrich Schütze. MEXA: Multilingual evaluation of English-centric LLMs via cross-lingual alignment. In *Findings of the Association for Computational Linguistics (ACL)*, 2025b. URL https://arxiv.org/abs/2410.05873.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019. URL https://arxiv.org/abs/1905.00414.

Jiaqi Mu, Suma Bhat, and Pramod Viswanath. All-but-the-top: Simple and effective postprocessing for word representations. In *International Conference on Learning Representations (ICLR)*, 2018. URL https://arxiv.org/abs/1702.01417.

Niklas Muennighoff, Qian Liu, Armel Zebaze, Qinkai Zheng, Binyuan Hui, Terry Yue Zhuo, Swayam Singh, Xiangru Tang, Leandro von Werra, and Shayne Longpre. Octopack: Instruction tuning code large language models. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2308.07124.

Aleksandar Petrov, Emanuele La Malfa, Philip H. S. Torr, and Adel Bibi. Language model tokenizers introduce unfairness between languages. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL https://arxiv.org/abs/2305.15425.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 8748–8763, 2021. URL https://arxiv.org/abs/2103.00020.

Paul Robert and Yves Escoufier. A unifying tool for linear multivariate statistical methods: The RV-coefficient. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 25(3):257–265, 1976. URL https://doi.org/10.2307/2347233.

Baptiste Rozière, Jonas Gehring, Fabian Gloeckle, et al. Code llama: Open foundation models for code. *arXiv preprint arXiv:2308.12950*, 2023. URL https://arxiv.org/abs/2308.12950.

Lisa Schut, Yarin Gal, and Sebastian Farquhar. Do multilingual LLMs think in english? In *ICLR 2025 Workshop on Building Trust in Language Models and Applications*, 2025. URL https://arxiv.org/abs/2502.15603.

Chris Wendler, Veniamin Veselovsky, Giovanni Monea, and Robert West. Do llamas work in English? on the latent language of multilingual transformers. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. URL https://aclanthology.org/2024.acl-long.820/.

A Additional Figures

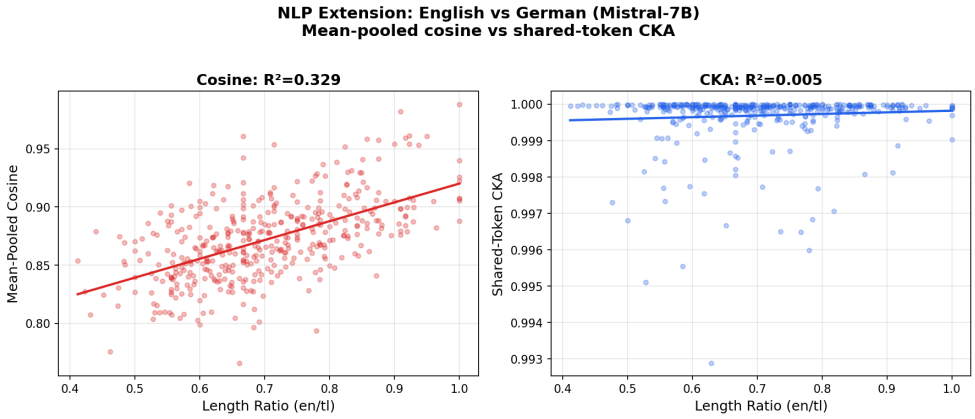


Figure 3: **NLP: English–German (Mistral-7B, n=428).** Cosine R^2=0.33; CKA R^2=0.005. Larger effect than French, consistent with German's longer tokenisation relative to English.

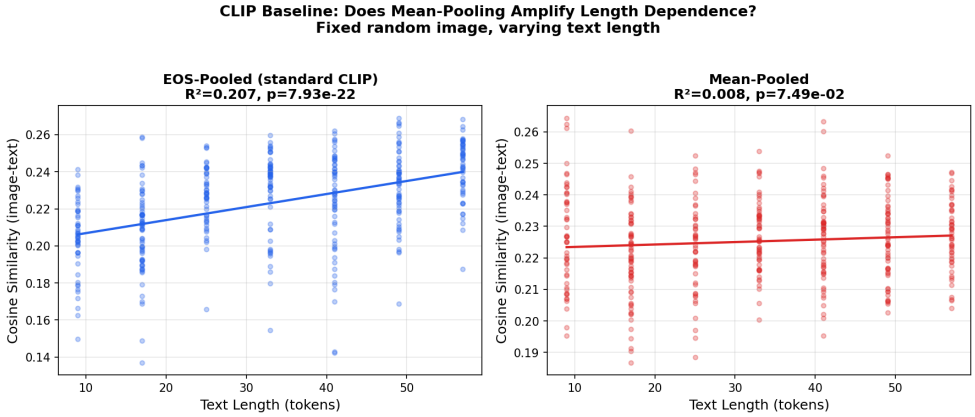


Figure 4: **CLIP ViT-B/32 (n=400).** EOS-pooled cosine (left): R^2=0.21. Mean-pooled cosine (right): R^2<0.01. Mean-pooling reduces the artifact because CLIP's contrastive head produces lower-anisotropy embeddings.

B Per-Pair Raw CKA Values

C AI Involvement Checklist

1. **Hypothesis development**

   Involvement level: A (Human-led; minimal AI use)

   Explanation: The core hypothesis—that mean-pooled cosine is length-dependent under anisotropy—arose from the human authors noticing anomalous R^2 values in preliminary cross-language experiments. The formal proposition, proof strategy, and framing were developed entirely by the human authors. AI tools were used only to check related-work coverage.

2. **Experimental design and implementation**

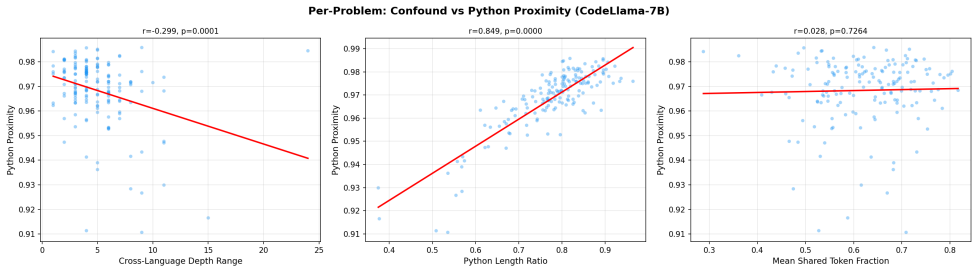


Figure 5: **Scatter: Python proximity vs. each confound (CodeLlama-7B).** Length ratio drives R^2=0.72; AST depth and shared tokens are flat once length is partialled out.

Table 3: Raw mean linear CKA at shared token positions ($\pm$ std across pairs).

Domain	Pair	Mean CKA	Std	n
Code	CodeLlama-7B: Py vs. {Java, JS, Go}	0.650	0.189	164
Code	CodeLlama-7B-Python: Py vs. {Java, JS, Go}	0.523	0.242	164
Code	CodeLlama-13B: Py vs. {Java, JS, Go}	0.997	0.001	164
Code	Qwen2.5-Coder-7B: Py vs. {Java, JS, Go}	0.913	0.063	164
NLP (FR)	Mistral-7B: EN vs. FR	0.9997	0.001	442
NLP (DE)	Mistral-7B: EN vs. DE	0.9997	$<$0.001	428

Involvement level: D (AI-executed; human-directed and validated)

Explanation: All analysis scripts were written by a large language model (Claude) under close human direction. The human authors specified every experimental decision: model selection, dataset choice, confound set, metric definitions, bootstrap parameters, and layer-averaging range. The human authors validated all outputs against expected theoretical behaviour before inclusion. The code is a direct expression of the human-designed experimental protocol.

3. **Analysis of data and interpretation of results**

Involvement level: A (Human-led; minimal AI use)

Explanation: All substantive interpretation—the Simpson's-paradox framing of AST depth, the sign-reversal as the central result, the cross-domain anisotropy explanation for CLIP, the genuine-convergence-vs-pivot-framing distinction—was developed by the human authors through multiple rounds of iteration. AI assistants were consulted only for sanity-checks on statistical terminology.

4. **Writing**

Involvement level: B (Human-led with AI editing assistance)

Explanation: The paper was drafted and iterated extensively by the human authors across many revision rounds. AI assistants (Claude) assisted with copyediting, LaTeX formatting, and a citation-verification pass that identified and corrected two fabricated references and three wrong-detail entries in the bibliography. All scientific framing, argumentation structure, and claim calibration are the human authors'.

D Reproducibility and Responsibility Checklist

1. **Claims**

Answer: **Yes**

Justification: All quantitative claims (Table 1, Table 2, Table 3) are supported by the corresponding CSV result files in the code repository. The R^2 values and β coefficients are reproduced by running `confound_regression.py`. The synthetic experiment (Fig. 1) requires no data and is fully self-contained.

2. **Limitations**

Answer: **Yes**

Justification: Section 6 (Limitations paragraph) discusses: adversarial failure modes of CKA (Davari et al., 2023), the shared-surface-form alignment selection effect, restriction to linear CKA, and the isotropic-noise approximation in Eq. (3).

3. **Theory assumptions and proofs**

Answer: **Yes**

Justification: Proposition 1 states assumptions explicitly ($d \gg 1$, i.i.d. draws, $\|\mu\|^2 = \Theta(d\mu_{\text{comp}}^2)$). The proof sketch in Section 3 covers numerator, denominator, and assembly. The isotropic-noise approximation is stated as a simplification; the qualitative monotonic dependence is what is empirically tested.

4. **Experimental result reproducibility**

Answer: **Yes**

Justification: All code, data-loading scripts, and analysis pipelines are available at `https://anonymous.4open.science/r/cka-length-invariance-E279/`. Models are publicly available on HuggingFace. Datasets (HumanEvalPack, WMT14/16) are publicly available. All random seeds and hyperparameters are fixed in the scripts.

5. **Open access to data and code**

Answer: **Yes**

Justification: Code is available at the anonymous repository above. All datasets (HumanEvalPack via HuggingFace, WMT via OPUS/Sacrebleu, Wikipedia via HuggingFace) are publicly available without access tokens.

6. **Experimental setting / details**

Answer: **Yes**

Justification: Section 4 specifies all model identifiers, dataset sizes, metric definitions, and confound variables. Layer averaging range (middle layers $n/4$ to $3n/4$) and alignment filter (≥ 3 shared tokens) are stated explicitly.

7. **Experiment statistical significance**

Answer: **Yes**

Justification: Bootstrap 95% CIs (B=5000) are reported for all R^2 values in Table 1. p-values are reported for the NLP results ($p<10^{-26}$ for cosine; p=0.69 for CKA on EN–FR). The sign reversal in β_{len} (Table 2) is the primary result and is large in magnitude ($+0.86 \rightarrow -0.37$), not marginal.

8. **Experiments compute resources**

Answer: **Yes**

Justification: All experiments ran on a single NVIDIA A100 (80 GB) GPU. The synthetic experiment (Sec. 3) requires only CPU. Total GPU time was approximately 6 hours. No distributed or proprietary compute was used.

9. **Research ethics**

Answer: **Yes**

Justification: All models and datasets used are publicly available. No human subjects, personal data, or sensitive content are involved. The paper critiques a widely-used metric; no fabricated or misleading experimental results are reported. Two bibliographic errors identified during citation verification (one fabricated arXiv ID, one incorrect title) were corrected before submission.

10. **Broader impacts**

Answer: **Yes**

Justification: The work identifies a methodological artifact in a widely-used metric. The primary societal impact is positive: more reliable measurement of cross-lingual representational structure. A potential negative impact is reputational harm to prior work; we mitigate by showing genuine convergence exists under CKA and framing the paper as a calibration of claims rather than a refutation of the underlying phenomenon.